Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/13/2019
PI Name:	Mesarovic, Sinisa Ph.D.		
Project Title:	Computational Framework for Capillary Flows		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pullman	**State:**	WA
Zip Code:	99164-2920	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-A: Use of the NASA Physical Sciences Informatics System – Appendix A
Start Date:	05/16/2016	**End Date:**	05/15/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Rame, Enrique	**Contact Phone:**	216-433-2842
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/15/2019 per NSSC information (Ed., 5/15/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AG57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Capillary (wetting) flows are the key element in many industrial and natural processes, such as soldering/ brazing, de-icing, organic liquid imbibing, etc. In some applications the requirement is simple: to enhance or limit the wetting at a given temperature. In other applications a precise control is required. Fluid transport systems based on capillary flows usually require a hardware that weighs less compared to other methods and operates efficiently in the low gravity environment. Thus, they are essential for space missions. Applications include fuel, life support, and thermal control. Each application uses specific geometry of the flow, which may be complex and is subject to changes and improvements. While the simple physical models designed for simple geometries have enormous impact on our understanding of the fluid physics, they cannot predict the flow in complex geometries, so that without a reliable computational model, the design process is dependent on repeated testing which, for low gravity conditions, is very expensive. We propose to use the experimental results obtained aboard the International Space Station (ISS): - The Capillary Flow Experiment (CFE), specifically the contact line (CL) and the vane gap (VG) configurations, and, - Capillary Channel Flow (CCF) experiments; For the purpose of: - Verification of the computational model for the capillary flows in complex geometries, and, - Enhancement of the continuum theory behind the computational model. The final objective is the development of the predictive computational model for capillary multi-phase flows in complex geometries capable of representing: - The intricacies of the triple line mobility, - Topological discontinuities such as: the breakup of the liquid film on the solid surface, droplet separation, and bubble adsorption under dynamic conditions. To be applicable to arbitrary geometries, the computational framework will be based on the finite element method. The mathematical formulation is based on the coupled Navier-Stokes and Cahn-Hilliard equations for multiphase flow in the volume, and the diffusive motion of the triple line in the solid surface. The proposed computational model will significantly reduce the need for iterative testing of various designs of fluid transport systems in low gravity. Another application significant to NASA is the wetting of liquid polymer of the carbon nanotubes (CNT) or CNT bundles during manufacturing of light-weight CNT-polymer composites for structural aerospace applications. The computational framework proposed here will be applicable to a range of industrial and natural phenomena and will enable understanding of natural processes and rational design of industrial processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The computational framework proposed here will be applicable to a range of industrial and natural phenomena and will enable understanding of natural processes and rational design of industrial processes. The capillary flow experiments aboard ISS feature micro-gravity conditions where capillarity dominates and represent unique opportunity for testing the model.
Task Progress:	At the end of the grant period, all tasks are near completion. What remains to be done is a few final computational runs and preparations of 3 additional manuscripts for submission to journals. This should be accomplished by the end of 2019. The graduate student who was supported mainly by this grant (Kumar) is expected to defend his dissertation by the end of 2019. The other graduate student who participated in this project (Lazaridis) is supported mainly by another grant and will complete his dissertation in 2020.
Bibliography Type:	Description: (Last Updated: 02/01/2021)
Abstracts for Journals and Proceedings	Mesarovic SDj, Kumar S, Krishna M. "Computational modeling of capillary flows in microgravity." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Articles in Other Journals or Periodicals	Kumar S, Krishna M, Mesarovic SDj. "Contact line dynamics in microgravity." in preparation as of May 2019. , May-2019
Articles in Other Journals or Periodicals	Kumar S, Krishna M, Mesarovic SDj. "Phase field simulations of capillary flow at corners." in preparation, as of May 2019. , May-2019
Books/Book Chapters	Mesarovic S. "Physical foundations of mesoscale continua." in "Mesoscale models: From micro-physics to macro-interpretation." Ed. S.Dj. Mesarovic, S. Forest, H.M. Zbib. Cham: Springer, 2019. p. 1-50. CISM International Centre for Mechanical Sciences book series, vol. 587. https://doi.org/10.1007/978-3-319-94186-8_1 , Jan-2019